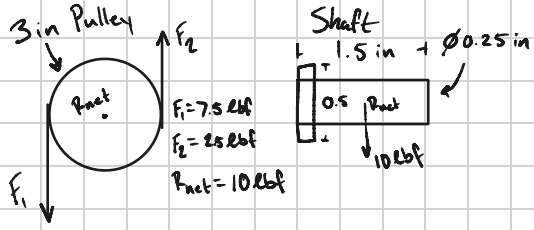
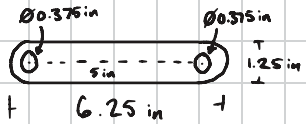
Given: $C = 5.0\,in$
$D = 3.0\,in$
$n = 1500\,rpm$
$F_i = 5.0\,lbf$
$\sigma_y = 471\,psi$
$SF = 3$





assuming torque load increase

$\sigma_{allow} = \frac{\sigma_y}{SF} = \frac{471}{3} = 157\,psi$ $F_1 = F_i + \Delta F = 5 + 2.5 = 7.5\,lbf$ $F_2 = 2F_i - F_1 = 2(5) - 7.5 = 2.5\,lbf$

$A = \frac{F_1}{\sigma_{allow}} = \frac{7.5}{157} = 0.0478\,in^2$ $d = \sqrt{\frac{4A}{\pi}} = \sqrt{\frac{4(0.0478)}{\pi}} = 0.247\,in$ $R = \frac{d}{2} = \frac{0.247}{2} = 0.1235\,in$

$L = 2C + \pi D = 2(5) + \pi(3) = 19.425\,in$